

Certificate Pricing

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1 Introduction

We assume that Bank XX issues a certificate (autocallable note) on the EURO STOXX 50 index. The product features a conditional digital coupon structure and an early termination clause (*Early Redemption*). To hedge its structured payoff, Bank XX enters a swap with an Investment Bank (I.B.). Under this agreement, at the start date t_0 the I.B. pays an upfront $X\%$ to Bank XX and throughout the life of the contract, Bank XX pays the floating leg — Euribor 3m — to the I.B., while the I.B. (Party B) mirrors the structured coupon payments back to Bank XX. The primary objectives of the project include:

- Calculating the Upfront premium ($X\%$) under mid-market conditions using the **Normal Inverse Gaussian (NIG)** and **Variance Gamma (VG)** models, calibrated to current market data. The valuation is carried out via two complementary approaches: the **Lewis analytical formula** and **Monte Carlo simulation**.
- Assessing the pricing error resulting from the use of the standard **Black model**, through the **Volatility Smile correction**.
- Extending the valuation to a three-year time horizon, analyzing the path-dependent nature of the contract.

2 Product Description

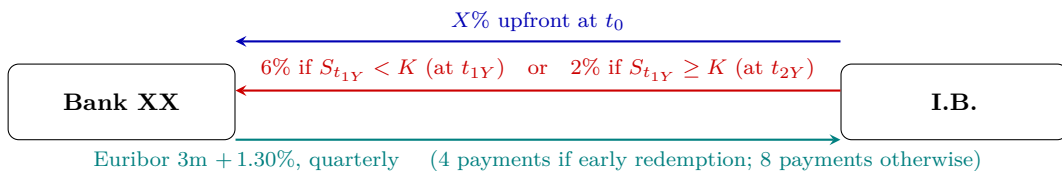
This case study considers the pricing of a structured certificate issued by Bank XX on February 15, 2008. The product is a 2-year note with a principal amount of 100 million EUR, issued at par, whose coupon payments are linked to the performance of the EURO STOXX 50 index. The coupon structure embeds an early redemption feature based on the index level at each annual reset date (set 2 business days prior to the respective coupon payment date). The payoff can be summarized as follows:

Condition at Year 1 Reset	Year 1 Outcome	Year 2 Outcome
$S_{t_{1Y}} < K = 3200$	Coupon 6% + Principal repayment	Nothing : Early redemption in t1
$S_{t_{1Y}} \geq K = 3200$	Coupon 0%	Coupon 2% + Principal repayment

Table 1: Payoff summary of the structured certificate issued by Bank XX.

Specifically, if the EURO STOXX 50 closes *below* the strike $K = 3200$ at the Year 1 reset date, the cumulative coupon accrual reaches the trigger level of 6%, the note is automatically redeemed at par and Bank XX pays a 6% coupon. Otherwise, no coupon is paid in Year 1 and the contract runs to maturity, where a final coupon of 2% is paid together with the principal.

To hedge its structured payoff, Bank XX enters a swap with an Investment Bank (I.B.). Under this agreement, at the start date t_0 the I.B. pays an upfront $X\%$ to Bank XX; throughout the life of the contract, Bank XX (Party A) pays the floating leg — Euribor 3m + 1.30% on a quarterly basis (Act/360 day count) — to the I.B., while the I.B. (Party B) mirrors the structured coupon payments back to Bank XX. The cash flow structure is illustrated below:



The goal of the exercise is to determine the fair value of $X\%$ under mid-market conditions, i.e., the upfront payment that makes the NPV of the swap equal to zero at inception. The fair value condition reads:

$$X + \text{NPV}_{\text{coupon}} = \text{NPV}_{\text{float}}, \quad (1)$$

where $\text{NPV}_{\text{coupon}}$ is the present value of the structured coupons paid by the I.B., and $\text{NPV}_{\text{float}}$ is the present value of the floating leg paid by Bank XX.

The floating leg NPV accounts for the path-dependency introduced by the early redemption clause: if $S_{t_{1Y}} < K$, only 4 quarterly payments are made; otherwise, all 8 are exchanged. Exploiting the independence of interest rates and equity dynamics, and using $B(t_0, t_i)$ for discount factors at the quarterly Euribor payment dates $t_1, t_2, \dots, t_8$ (with $t_4 \equiv t_{1Y}$ and $t_8 \equiv t_{2Y}$) and $\delta(t_{i-1}, t_i)$ for the corresponding Act/360 day-count fractions (Euribor payments simplify via a telescoping sum argument) we get:

$$\begin{aligned} \text{NPV}_{\text{float}} = & 1 - B(t_0, t_{1Y}) + \underbrace{1.30\% \sum_{i=1}^4 \delta(t_{i-1}, t_i) B(t_0, t_i)}_{\text{Year 1 (always paid)}} \\ & + \underbrace{[B(t_0, t_{1Y}) - B(t_0, t_{2Y})]}_{\text{telescopic sum, Year 2}} \cdot \mathbb{P}(S_{t_{1Y}} \leq K) \\ & + 1.30\% \sum_{i=5}^8 \delta(t_{i-1}, t_i) B(t_0, t_i) \cdot \mathbb{P}(S_{t_{1Y}} \leq K), \end{aligned} \quad (2)$$

while the coupon leg NPV is:

$$\text{NPV}_{\text{coupon}} = 6\% \cdot B(t_0, t_{1Y}) \delta^*(t_0, t_{1Y}) \mathbb{P}(S_{t_{1Y}} \leq K) + 2\% \cdot B(t_0, t_{2Y}) \delta^*(t_{1Y}, t_{2Y}) \mathbb{P}(S_{t_{1Y}} > K), \quad (3)$$

where $\delta^*(\cdot, \cdot)$ denotes the day-count fraction computed under the 30/360 European convention, as specified in the coupon leg of the swap termsheet.

2.1 Market Data and Discount Curve

The valuation is performed as of **15 February 2008**, using EUR interest rate market data sourced from the interbank market on that date. The discount curve is bootstrapped from three market segments, in order of increasing maturity:

- **Deposits** (short end): overnight to 3-month instruments, discounted via the standard money-market formula;
- **Futures** (medium term): Euribor futures contracts, used to extract forward discount factors via the futures strip;
- **Interest Rate Swaps** (long end): plain vanilla EUR fixed-for-floating swaps versus Euribor 3m, bootstrapped iteratively using the par-swap pricing condition.

At each step, mid-market rates (average of bid and ask) are used. Discount factors at arbitrary dates are then obtained by linear interpolation of the continuously compounded zero rates under the ACT/365 day-count convention. All coupon and Euribor payment dates are adjusted to business days under the *Modified Following* convention.

3 NIG model valuation

The exercise required pricing the certificate under the Normal Inverse Gaussian (NIG) model. In this framework, the log-returns of the underlying asset are modeled using a Normal Mean-Variance Mixture, where the dynamics can be expressed as:

$$f_t = \sqrt{t - t_0} \sigma \sqrt{G} g - \left(\frac{1}{2} + \eta \right) (t - t_0) \sigma^2 G - \ln \mathcal{L}[\eta] \quad (4)$$

In this equation, $g \sim \mathcal{N}(0, 1)$ represents a standard normal distribution, while G is an Inverse Gaussian subordinator with parameters $\mu = t$ and $\lambda = t^2/\kappa$ and the laplace exponent defined as $\mathcal{L}(u) = \frac{1 - \sqrt{1 + 2\kappa u}}{\kappa}$.

The discount factors used for the valuation were derived from the bootstrap curve constructed in previous assignments. The NIG model parameters were recalibrated following the methodology established in Laboratory 4, yielding the following results:

To price the certificate, two methodologies were implemented: an analytical approach using Lewis formula and a numerical approach via Monte Carlo simulation.

Parameter	Value
σ	0.1742
κ	0.4526
η	10.2141

Table 2: NIG Calibration Results

3.0.1 Analytical Approach

Following the Lewis framework [see 3.12 in Lewis paper], the pricing of the digital component was performed using the following formulas:

$$\Pi_1 = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{iuk} \phi_T(u-i)}{iu} \right] du, \quad \Pi_2 = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \operatorname{Re} \left[\frac{e^{iuk} \phi_T(u)}{iu} \right] du \quad (5)$$

Specifically, Π_2 represents the risk-neutral probability $\Pr(S_T > K)$. Hence, the Digital Put probability was calculated as $1 - \Pi_2$. The integration was performed using the `quadgk` adaptive quadrature. The analytical results are summarized below:

Item	Value (%)
Digital Put Prob.	24.7135%
NPV Floating Leg	8.6807%
NPV Coupon Leg	2.8211%
X Upfront	5.8595%

Table 3: Analytical Valuation Results (Lewis)

3.0.2 Monte Carlo Approach

The Monte Carlo valuation was conducted using 10^7 simulations. To capture the path-dependency of the contract, we simulated the terminal value of the forward $F(t, t_1)$ at the first reset date, verifying for each path whether the digital condition was met. The payoff logic was then structured using boolean indicators to manage the conditional cash flows and the early redemption trigger. This binary mapping allowed for an efficient calculation of both the expected values and the associated standard error. The diagnostics for the simulation are as follows:

Item	Value (%)
Digital Put Prob.	24.7277%
NPV Floating Leg	8.6800%
NPV Coupon Leg	2.8217%
Standard Error	0.12 bps
X Upfront	5.8583%

Table 4: Monte Carlo Simulation Results (10^7 simulations)

3.0.3 Comparison and Conclusions

A final comparison between the analytical Lewis approach and the Monte Carlo simulation demonstrates the high degree of accuracy of the numerical model.

Method	Digital Put Prob.	Upfront (%)	Upfront Value (100M)
Lewis	24.7135%	5.8595%	5,859,500 EUR
Monte Carlo	24.7277%	5.8583%	5,858,300 EUR

Table 5: Comparison of Results

The numerical results closely match the theoretical risk neutral values. The minor discrepancy is well within the limits defined by the Monte Carlo standard error, confirming that the simulation correctly captures the NIG dynamics and the certificate's path-dependent features.

4 VG Model Valuation

As an alternative to the NIG model, we consider the Variance Gamma (VG) process. Both models belong to the family of Normal Mean-Variance Mixtures, and share the same general structure for the log-return increment. The key difference between the two models lies in the choice of the subordinator: in the VG model G follows a *Gamma* distribution with shape $a = t/\kappa$ and scale $b = \kappa$, and the Laplace exponent is $\mathcal{L}(u) = -\frac{1}{\kappa} \log(1 + \kappa u)$. Calibrating as before we obtained the following parameters:

Parameter	Value
σ	0.2115
κ	0.6674
η	4.2850

Table 6: VG Calibration Results.

Compared to the NIG calibration, the VG model yields a higher σ and κ and a lower η , reflecting the different tail behaviour of the Gamma subordinator with respect to the Inverse Gaussian one.

4.1 Monte Carlo Valuation

The Monte Carlo simulation follows exactly the same structure described for the NIG model in Section 2.1.2, with the only difference being the simulation of the subordinator G : instead of drawing from an Inverse Gaussian distribution, increments are drawn from a Gamma distribution with shape $a = \Delta t/\kappa$ and scale $b = \kappa$. The results with 10^7 simulations are:

Item	Value
Digital Put Prob.	25.0617%
NPV Floating Leg	8.6600%
NPV Coupon Leg	2.8300%
Standard Error	0.12 bps
X Upfront	5.8296%

Table 7: VG Monte Carlo Results (10^7 simulations).

Model	Digital Put Prob.	Upfront (%)	Upfront Value (100M)
NIG	24.7277%	5.8583%	5,858,300 EUR
VG	25.0617%	5.8296%	5,829,600 EUR

Table 8: Comparison between NIG and VG models.

5 Distribution Comparison: NIG vs. VG Subordinators

To gain a deeper understanding of how the two models interpret the underlying dynamics, we analyzed the probability density functions of the subordinator G . By analyzing G , we can observe how the models distribute volatility and price jumps differently across the same time horizon.

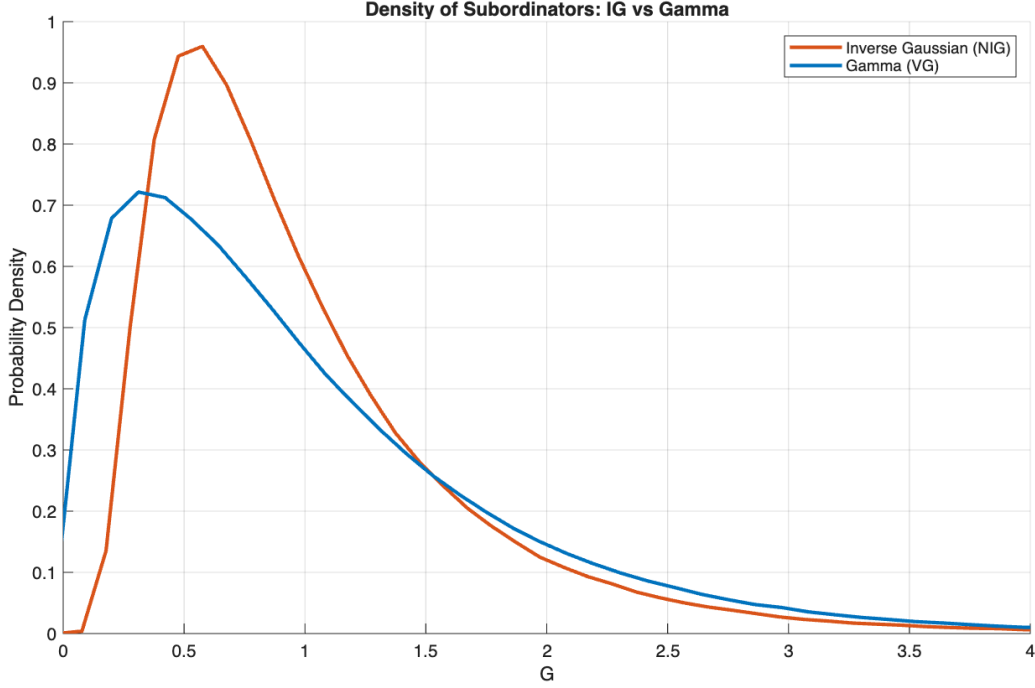


Figure 1: Probability density function comparison between the Inverse Gaussian (NIG) and Gamma (VG) subordinators.

The plot in Figure 1 illustrates the density of the subordinators for the NIG and VG models. While both share the same mean, the Inverse Gaussian (NIG) exhibits a higher concentration around the mean with a specific semi-heavy tail decay. This characteristic allows the NIG model to capture more extreme market shocks (fat tails) compared to the standard normal distribution. Conversely, the Gamma distribution (VG) exhibits more mass near zero, highlighting the different way these models distribute price jumps and instantaneous variance over time.

6 Black 76

The key quantity driving the entire pricing is the risk-neutral probability $\mathbb{P}(S_{t_{1Y}} \leq K)$, which enters both the coupon leg and the floating leg. Since the payoff depends on the index level only at a single reset date t_{1Y} , the problem reduces to pricing a digital put option on the EURO STOXX 50 with strike $K = 3200$ and maturity T_1 . This observation opens the door to an alternative, simpler model: the Black-76 framework.

Plain Black-76 Pricing

Under the Black-76 model, the forward price $F(t_0, t_{1Y})$ is assumed to follow a log-normal dynamics with constant volatility σ . The risk-neutral probability of the digital put is given by:

$$\mathbb{P}_{\text{Black}}(S_{t_{1Y}} < K) = \mathcal{N}(-d_2), \quad (6)$$

where

$$d_1 = \frac{\log(F/K) + \frac{1}{2}\sigma^2 T_1}{\sigma\sqrt{T_1}}, \quad d_2 = d_1 - \sigma\sqrt{T_1}, \quad (7)$$

and $\mathcal{N}(\cdot)$ denotes the standard normal CDF. The implied volatility σ is read off the market surface at the relevant strike $K = 3200$ and maturity $T_1 \approx 1$ year via spline interpolation. Once $\mathbb{P}_{\text{Black}}$ is obtained, the upfront $X\%$ follows from the same pricing equations derived above for the NIG model.

Smile Correction: Managing Digital Risk

The plain Black formula ignores the implied volatility smile: it prices the digital option as if σ were flat across strikes, whereas in practice $\sigma = \sigma(K)$ varies. The correct price of the digital put, obtained by differentiating the market put price $p[K, \sigma(K)]$ with respect to K , reads:

$$dp = +\frac{d}{dK} p[K, \sigma(K)] = +\frac{\partial p}{\partial K} \Big|_{\sigma} + \frac{\partial \sigma}{\partial K} \cdot \frac{\partial c}{\partial \sigma}, \quad (8)$$

where the first term is the standard Black digital (the Black term) and the second is the *slope impact*, which corrects for the skew of the implied volatility surface. Rearranging, the smile-adjusted probability is:

$$\mathbb{P}_{\text{smile}}(S_{t_{1Y}} \leq K) = \mathcal{N}(-d_2) + \frac{\partial \sigma}{\partial K} \cdot \mathcal{V}_{\text{Black}}, \quad (9)$$

where $\mathcal{V}_{\text{Black}} = F\sqrt{T_1} \phi(d_1)$ is the Black-76 undiscounted vega and $\phi(\cdot)$ is the standard normal PDF. The smile slope is estimated numerically via a central finite difference on the market surface:

$$\frac{\partial \sigma}{\partial K} \approx \frac{\sigma(K + \varepsilon) - \sigma(K - \varepsilon)}{2\varepsilon}, \quad (10)$$

with $\varepsilon = 1$ index point. The smile-corrected upfront $X_{\text{smile}}\%$ is then computed by substituting $\mathbb{P}_{\text{smile}}$ into the pricing equations in place of $\mathbb{P}_{\text{Black}}$.

Note that the Black model, with or without smile correction, is applicable here precisely because the payoff depends on the index at a *single* observation date, as we will see in section 2.3.

Method	Upfront (%)
Black-76 plain	5.2142%
Black-76 smile-adjusted	5.8275%
Difference	0.6133%

Table 9: Black-76 pricing results: plain vs. smile-adjusted (digital risk correction).

The smile correction has a significant impact of approximately 61 basis points, highlighting the importance of accounting for the slope of the implied volatility surface when pricing digital-type payoffs. The smile-adjusted Black result of 5.8275% is moreover in close agreement with both the NIG (5.8595%) and VG (5.8296%) estimates, confirming the consistency of the three approaches.

7 Extended Certificate Valuation

We now consider a three-year extension of the same certificate, where the Year 1 payoff structure is replicated also in Year 2, and early redemption can occur at the end of either the first or the second year. The payoff structure becomes:

Condition	Year 1 Outcome	Year 2 Outcome	Year 3 Outcome
$S_{t_{1Y}} < K$	Coupon 6% + Principal	Early redemption	—
$S_{t_{1Y}} \geq K, S_{t_{2Y}} < K$	Coupon 0%	Coupon 6% + Principal	Early redemption
$S_{t_{1Y}} \geq K, S_{t_{2Y}} \geq K$	Coupon 0%	Coupon 0%	Coupon 2% + Principal

Table 10: Payoff summary of the three-year structured certificate.

The floating leg paid by Bank XX now spans up to three years: 4 quarterly payments are made if early redemption occurs at t_{1Y} , 8 payments if the contract survives to t_{2Y} but redeems early there, and all 12 payments if the contract runs to full maturity t_{3Y} . The NPV of the floating leg is therefore:

$$\begin{aligned} \text{NPV}_{\text{float}} = & 1 - B(t_0, t_{1Y}) + \underbrace{1.30\% \sum_{i=1}^4 \delta(t_{i-1}, t_i) B(t_0, t_i)}_{\text{Year 1 (always paid)}} \\ & + [B(t_0, t_{1Y}) - B(t_0, t_{2Y})] \cdot \mathbb{P}(S_{t_{1Y}} > K) \\ & + 1.30\% \sum_{i=5}^8 \delta(t_{i-1}, t_i) B(t_0, t_i) \cdot \mathbb{P}(S_{t_{1Y}} > K) \\ & + [B(t_0, t_{2Y}) - B(t_0, t_{3Y})] \cdot \mathbb{P}(S_{t_{1Y}} > K, S_{t_{2Y}} > K) \\ & + 1.30\% \sum_{i=9}^{12} \delta(t_{i-1}, t_i) B(t_0, t_i) \cdot \mathbb{P}(S_{t_{1Y}} > K, S_{t_{2Y}} > K), \end{aligned} \quad (11)$$

where the Euribor contributions in Years 2 and 3 again collapse to differences of discount factors via the telescoping identity. The coupon leg NPV becomes:

$$\begin{aligned}\text{NPV}_{\text{coupon}} = & 6\% \cdot B(t_0, t_{1Y}) \delta^*(t_0, t_{1Y}) \mathbb{P}(S_{t_{1Y}} \leq K) \\ & + 6\% \cdot B(t_0, t_{2Y}) \delta^*(t_{1Y}, t_{2Y}) \mathbb{P}(S_{t_{1Y}} > K, S_{t_{2Y}} \leq K) \\ & + 2\% \cdot B(t_0, t_{3Y}) \delta^*(t_{2Y}, t_{3Y}) \mathbb{P}(S_{t_{1Y}} > K, S_{t_{2Y}} > K).\end{aligned}\tag{12}$$

In this three-year setting, the payoff depends on the *joint* distribution of $(S_{t_{1Y}}, S_{t_{2Y}})$, i.e. on the index level at two distinct observation dates. This rules out the use of the Black-76 model (with or without smile correction), which only provides the marginal distribution at a single date and cannot capture the path-dependency across multiple reset dates.

Conversely, the Monte Carlo method remains a robust tool for pricing such path-dependent structures. This is due to the fundamental properties of the NIG and VG processes, which belong to the class of Lévy processes characterized by independent and stationary increments. This property allows for a step-by-step simulation of the price path: the evolution from t_1 to t_2 can be generated independently of the previous path, provided the starting value is set to the simulated price at t_1 . Although the model parameters were initially calibrated to the one-year volatility smile, they are assumed to be constant across the extended maturity as per the exercise hypothesis.

The simulation was implemented using 10^7 paths, incorporating a system of boolean indicator vectors to manage the survival of each scenario. Only paths that did not trigger early redemption at the first reset date contributed to the subsequent year's cash flows. The results for both the NIG and VG models are displayed in the tables below.

Item	Value (%)
Digital Put Prob. (Yr 1)	24.7277%
Digital Put Prob. (Yr 2 Survived)	20.1124%
NPV Floating Leg	11.5200%
NPV Coupon Leg	3.3500%
Monte Carlo Standard Error	0.18 bps
X Upfront	8.1761%

Table 11: Monte Carlo Results: NIG Model (3-Year Maturity)

Item	Value (%)
Digital Put Prob. (Yr 1)	25.0617%
Digital Put Prob. (Yr 2 Survived)	20.2381%
NPV Floating Leg	11.4900%
NPV Coupon Leg	3.3600%
Monte Carlo Standard Error	0.19 bps
X Upfront	8.1291%

Table 12: Monte Carlo Results: VG Model (3-Year Maturity)